Appendix

A Experimental Setup

We follow an experimental setup similar to the one used in previous work [6, 17, 26], where we combine the Natural Questions (NQ) and HotpotQA datasets for training. We use Qwen2.5-3B and Qwen2.5-3B Instruct as backbone to our model and train using group relative policy optimization (GRPO) using the *verl* framework. We apply GRPO with a group size of 5, a rollout temperature of 1.0, a training batch size of 512, and a validation batch size of 256. We set a maximum prompt length to 4096 tokens, a response length to 500 tokens, and an observation length of 1200 tokens. We employ a learning rate of 1×10^{-6} and a KL divergence coefficient (kl_loss_coef) of 0.001. The model is trained using 4 NVIDIA H100 GPUs.

			RL				
Method	**Evidence**	**SFT**	**Rewards**	**Type**	**Training data**	**Decomp.**	**Interm.**
Direct Inference	–	–	–	–	–	–	–
CoT	–	–	–	–	–	–	–
SFT	–	✓	–	–	–	–	–
RAG	✓	–	–	–	–	–	–
Search-o1	✓	–	–	–	–	–	–
R1-base	–	–	EM	global	NQ+HotpotQA	–	–
Search-R1	✓	–	EM	global	NQ+HotpotQA	–	–
ZeroSearch	–	–	EM	global	NQ+HotpotQA	–	–
StepSearch	✓	–	EM	global	MuSiQue	✓	–
R-Search	✓	–	EM+evidence q.	global	2wiki	–	–
InForage	✓	✓	EM+gain+eff.	global	NQ+HotpotQA	–	–
O^2-Searcher	✓	✓	EM+fact+div.	global	NQ+HotpotQA	–	–
SubSearch	✓	–	EM+ans+dec.	**1g+2i**	NQ+HotpotQA	✓	✓

Table 2: Comparison of search agent methods. ✓/–: present/absent. *Evidence*: retrieved external documents. *Decomp.*: query decomposition. *Interm.*: subquery-level intermediate reward. **1g+2i**: 1 global + 2 intermediate.

B Format Analysis

Figure 3 shows that adding formatting is vital to avoid early model collapse. While for non-decomposed generations the model seems more stable during training, adding parallel decomposition increases the complexity of the generation and therefore makes it

more volatile. Hence, we use a combination of outcome-only, intermediate, and format rewards for stable and efficient training of SubSearch.

$$r_{\text{format}} = \begin{cases} 0 & \text{if } f_{\text{format}} = \text{False} \wedge f_{\text{retrieval}} = \text{False} \\ \lambda_{\text{structure}} & \text{if } f_{\text{format}} = \text{True} \wedge f_{\text{retrieval}} = \text{False} \\ \lambda_{\text{structure}} + \lambda_{\text{retrieval}} & \text{if } f_{\text{format}} = \text{True} \wedge f_{\text{retrieval}} = \text{True}, \end{cases} \tag{8}$$

where $\lambda_{\text{structure}}$ and $\lambda_{\text{retrieval}}$ are set values.

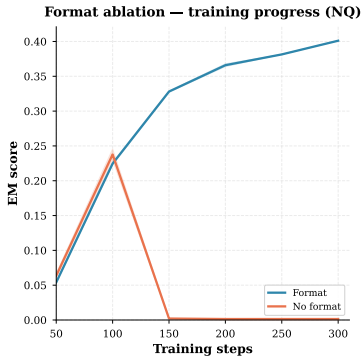


Fig. 3: Training progress of Qwen2.5-3B-base on NQ with GRPO and EM as reward, with and without using a format signal.

C Adaptive beta

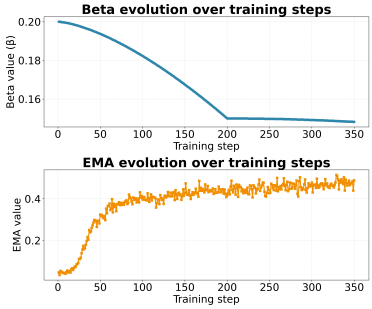


Fig. 4: Adaptive beta (β) evolution over training steps. A higher value highlights more weight to the intermediate rewards, while a lower value shows the model becomes better at giving a correct answer and therefore gives priority to the binary outcome reward. Higher EMA highlights giving more weight to recent reasoning traces compared to older ones.

D Query Decomposition and Rewards

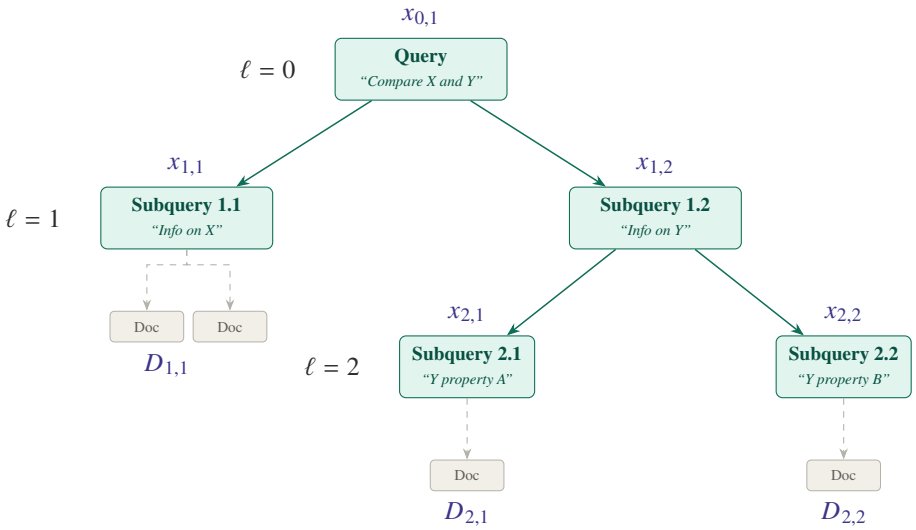


Fig. 5: SubSearch notation for query decomposition and intermediate rewards computation. l denotes the decomposition layer (root = 0), i denotes the index within each layer, $x_{l,i}$ represents a subquery at layer l and index i, and $D_{l,i}$ represents the retrieved documents for subquery $x_{l,i}$.

E Examples and Prompts

We include three case studies to illustrate how SubSearch successfully answer questions (Table 4, 5, 6) and one case where SubSearch is not successful due to a failure in decomposition (Table 7).

Answer the given question. You must conduct reasoning inside **<think>** and **</think>** first every time you get new information. After reasoning, if you find you lack some knowledge, you can call a search engine by **<search>** *query* **</search>**, and it will return the top searched results between **<information>** and **</information>**.

If the original query is complex or involves multiple parts, you are encouraged to **decompose** it into at most 3 smaller sub-questions, separated by ##. For example: **<search>** sub-question 1 ## sub-question 2 **</search>** and it will return the top searched results between **<information>** documents sub-question 1 ## documents sub-question 2 **</information>**.

You can search as many times as you want. Only **decompose** when the question has multiple independent parts (e.g., different entities, aspects, or comparisons). Do not decompose questions that do not need it.

If you find no further external knowledge needed, you can directly provide the answer inside **<answer>** and **</answer>** without detailed illustrations. For example, **<answer>** Beijing **</answer>**.

Question: {question}

Table 3: Prompt template for SubSearch. The question is appended during training and inference.

Question: Which bank has more branches, China CITIC Bank or UniCredit?

Ground Truth: UniCredit

<think> First, I need to find information about China CITIC Bank. **</think>**
<search> how many branches does UniCredit have bank **</search>**
<information> Doc 1(Title: "UniCredit Bank Romania") UniCredit Bank Romania UniCredit Bank is a leading European Bank, member of UniCredit Group, with a network of 8.500 branches in 17 European countries and it is present in another 50 international markets Being one of the top 5 banks on the Romanian market, UniCredit Bank has 208 branches and almost 3000 employees. At the end of 2015, the bank had total assets in amount of 34,6 billion RON and included approximately 600.000 active customers. Among the offered services are: financial solutions for individuals, SMEs, companies and freelancers. UniCredit Bank is recognized for specializing in the real estate market and.. Doc 2(Title: UniCredit) UniCredit UniCredit S.p.A. is an Italian global banking and financial services company. Its network spans 50 markets in 17 countries, with more than 8,500 branches and over 147,000 employees. Its strategic position in Western and Eastern Europe gives the group one of the region's highest market shares. The company is a component of the Euro Stoxx 50 stock market index. UniCredit Group was the outcome of the 1998 merger of several Italian banking groups, which the majority one were Unicredito (banks from Turin, Verona and Treviso) and Credito Italiano (consists of Rolo Banca, Banca Popolare di Rieti), hence the name.. Doc 3(Title: UniCredit) subsidiary Bank Austria). Until 2018 UniCredit was one of the UEFA Champions League's main sponsors. UniCredit UniCredit S.p.A. is an Italian global banking and financial services company. Its network spans 50 markets in 17 countries, with more than 8,500 branches and over 147,000 employees. Its strategic position in Western and Eastern Europe gives the group one of the region's highest market shares. The company is a component of the Euro Stoxx 50 stock market index. UniCredit Group was the outcome of the 1998 merger of several Italian banking groups, which the majority one were.. **</information>**
<think> Based on the information found, I need to compare the number of branches of China CITIC Bank and UniCredit. **</think>**
<search> how many branches does China CITIC Bank have **</search>**
<information> Doc 1(Title: "China CITIC Bank") China CITIC Bank China CITIC Bank () is China's seventh-largest lender in terms of total assets. It was known as CITIC Industrial Bank until it changed its name in August 2005. China CITIC Bank, established in 1987, is a nationally comprehensive and internationally oriented commercial bank. The bank operates in almost 130 countries, and maintains a strong foothold on the mainland banking industry. The bank operates 78 branches in the mainland, and 622 sub-branches, located in economically developed regions of China. In total, there are 773 branch offices in China, as of Q4 2011. In 1984, the chairman of CITIC.. Doc 2(Title: "China CITIC Bank") financing services for small enterprises, trading services, cash management, asset management, international settlement and trade financing, capital market business, including foreign exchange businesses, bond businesses, trading of financing and derivative products, financial service platforms and financial solutions. China CITIC Bank China CITIC Bank () is China's seventh-largest lender in terms of total assets. It was known as CITIC Industrial Bank until it changed its name in August 2005. China CITIC Bank, established in 1987, is a nationally comprehensive and internationally oriented commercial bank. The bank operates in almost 130 countries, and maintains a strong foothold on the mainland banking industry.. Doc 3(Title: UniCredit) the bank was also relocated from Genoa (inherited from Credito Italiano) to 17 via Minghetti, Rome. In the same year, two more acquisitions were carried out: ATF Bank, which ranks fifth out of domestic banks in Kazakhstan with 154 branches, and Ukrsotsbank, a universal bank in Ukraine. With these two banks the Group extended its operations in this area to 19 countries (including Central Asia). However, in November 2012, Kazakh government sources declared UniCredit is in talks with Kazakh investors over the sale of a controlling stake in ATF Bank. In 2010, UniCredit S.p.A. absorbed its Italian banking subsidiaries: UniCredit.. **</information>**
<think> Based on the information, China CITIC Bank has 773 branches, while UniCredit has 8,500 branches. **</think>**
<answer> UniCredit **</answer>**

Table 4: SubSearch case study 1 (successful): SubSearch can decompose the initial information need into parallel subqueries and aggregate the retrieved documents to answer correctly.

Question: Which team did Joe Buck's father broadcast for?
Ground Truth: St. Louis Cardinals
<think> First, I need to find information about Joe Buck. **</think>** **<search>** who is joe buck father broadcast **</search>** **<information>** Doc 1(Title: "Joe Buck") but rather to vocal cord paralysis likely caused by anesthesia used during multiple hair transplantation procedures. From 1993 to 2011, Buck was married to Ann Archambault, with whom he had two children. He married NFL Network and now ESPN reporter and former Bronco cheerleader Michelle Beisner on April 12, 2014. On April 26, 2018 they welcomed twin sons Wyatt Joseph and Blake Andrew. Joe Buck Joseph Francis Buck (born April 25, 1969) is an American sportscaster and the son of sportscaster Jack Buck. He has won numerous Sports Emmy Awards for his work with Fox Sports, including his roles as.. Doc 2(Title: "Joe Buck") the team's broadcasting crew. In 1994, Buck was hired by Fox, and at the age of 25 became the youngest man ever to announce a regular slate of National Football League games on network television. In , he was named Fox's lead play-by-play voice for Major League Baseball, teaming with Tim McCarver, who had previously worked with his father on CBS. That year, he became the youngest man to do a national broadcast (for all nine innings and games, as a network employee as opposed to simply being a representative of one of the participating teams) for a World Series.. Doc 3(Title: "Joe Buck") Joe Buck Joseph Francis Buck (born April 25, 1969) is an American sportscaster and the son of sportscaster Jack Buck. He has won numerous Sports Emmy Awards for his work with Fox Sports, including his roles as lead play-by-play announcer for the network's National Football League and Major League Baseball coverage, and is a three-time recipient of the National Sportscaster of the Year award. Since 1996, he has served as the play-by-play announcer for the World Series, each year, with the exceptions of 1997 and 1999. Since 2015, he's hosted ""Undeniable with Joe Buck"" on Audience Network. Buck was born**</information>** **<think>** I now need to find which team his father broadcast for. **</think>** **<search>** who did jack buck broadcast for **</search>** **<information>** Doc 1(Title: "Jack Buck") a railroad accountant who commuted weekly to New Jersey. From an early age, Buck dreamed of becoming a sports announcer with his early exposure to sports broadcasting coming from listening to Boston Red Sox baseball games announced by Fred Hoey. Part of his childhood coincided with the Great Depression, and Buck remembered his family sometimes using a metal slug to keep a coin-operated gas meter going during the winter to provide heat for their home. In 1939, his family moved to Cleveland, Ohio to join their father, who had a job with the Erie Railroad. Soon after though, Buck's father.. Doc 2(Title: "Jack Buck") after vomiting on a plane during a flight after a game, and was out for a considerable amount of time. While Verne Lundquist replaced Summerall on games with lead analyst John Madden, Buck (who was at the time the network's lead Major League Baseball announcer) filled in for Lundquist, teaming with Dan Fouts to call two games (both of which coincidentally featured the Cardinals, who had moved from St. Louis to Arizona by that time). While much better known for his baseball and football commentary, Jack Buck was also the original voice of the St. Louis Blues of the National.. Doc 3(Title: "Jack Buck") Jack Buck John Francis ""Jack"" Buck (August 21, 1924 – June 18, 2002) was an American sportscaster, best known for his work announcing Major League Baseball games of the St. Louis Cardinals. His play-by-play work earned him recognition from numerous Halls of Fame, such as the National Baseball Hall of Fame, the Pro Football Hall of Fame, and the National Radio Hall of Fame. He has also been inducted as a member of the St. Louis Cardinals Hall of Fame Museum. Buck was born in Holyoke, Massachusetts, the third of seven children of Earle and Kathleen Buck.. **</information>** **<think>** Therefore, Joe Buck's father broadcast for the St. Louis Cardinals. **</think>** **<answer>** St. Louis Cardinals **</answer>**

Table 5: SubSearch case study 2 (successful): SubSearch can decompose the initial information need into sequential subqueries (we need the answer of one to formulate the other) and aggregate the retrieved documents to answer correctly.

Question: In what year did the man who shot Chris Stockley, of The Dingoes, die?

Ground Truth: 1987

<think> First, I need to find information about Chris Stockley. **</think>**
<search> when did Chris Stockley of The Dingoes die **</search>**
<information> Doc 1(Title: "The Dingoes") The Dingoes The Dingoes is an Australian country rock band initially active from 1973 to 1979, formed in Melbourne which relocated to the United States from 1976. Most stable line-up was John Bois on bass guitar, John Lee on drums, Broderick Smith on vocals and harmonica, Chris Stockley on guitar, and Kerryn Tolhurst on guitar. Mal Logan (who provided keyboards on the first LP) on keyboards joined after Stockley was hospitalised when shot in the stomach by Melbourne drug-dealer, Dennis Allen, who was attempting to gate crash a party. The Dingoes' debut single, ""Way Out West"", was released in November.. Doc 2(Title: "The Dingoes") and Tolhurst although Bois' book confirms that the song was written by Tolhurst alone. A week before the single was released Stockley received a serious gunshot wound during an incident at a party in Melbourne that resulted in a two-month stay in hospital, initially described as an 'accidental shooting', according to music historian, Ian McFarlane's ""Encyclopedia of Australian Rock and Pop"",...**</information>**
<think> Now, I need to find information about the man who shot him. **</think>**
<search> who shot Chris Stockley of The Dingoes **</search>**
<information> Doc 1(Title: "The Dingoes") and Tolhurst although Bois' book confirms that the song was written by Tolhurst alone. A week before the single was released Stockley received a serious gunshot wound during an incident at a party in Melbourne that resulted in a two-month stay in hospital, initially described as an 'accidental shooting', according to music historian, Ian McFarlane's ""Encyclopedia of Australian Rock and Pop"", Stockley was shot by notorious drug dealer Dennis Allen, who was trying to gate crash the party. An eight-hour benefit concert was held for Stockley on 4 November 1973 at Leggett's Ballroom, Greville Street, Prahran. While recuperating, Stockley was.. Doc 2(Title: "The Dingoes") The Dingoes The Dingoes is an Australian country rock band initially active from 1973 to 1979, formed in Melbourne which relocated to the United States from 1976. Most stable line-up was John Bois on bass guitar, John Lee on drums, Broderick Smith on vocals and harmonica, Chris Stockley on guitar, and Kerryn Tolhurst on guitar... **</information>**
<think> I now need to find when Dennis Allen died. **</think>**
<search> when did Dennis Allen die **</search>**
<information> Doc 1(Title: "The Dingoes") and Tolhurst although Bois' book confirms that the song was written by Tolhurst alone. A week before the single was released Stockley received a serious gunshot wound during an incident at a party in Melbourne that resulted in a two-month stay in hospital, initially described as an 'accidental shooting', according to music historian, Ian McFarlane's ""Encyclopedia of Australian Rock and Pop"", Stockley was shot by notorious drug dealer Dennis Allen, who was trying to gate crash the party. An eight-hour benefit concert was held for Stockley on 4 November 1973 at Leggett's Ballroom, Greville Street, Prahran. While recuperating, Stockley was.. Doc 2(Title: "Dennis Allen (criminal)") in 1985. One victim who survived was guitarist Chris Stockley of The Dingoes, whom Allen shot in the stomach while attempting to gatecrash a party. Allen received a ten-year prison sentence for rape during the 1970s. It is also reported that he was a major drug dealer in the Richmond and South Yarra areas of Melbourne during the 1980s. New South Wales Police Detective-Sergeant Roger Rogerson was convicted of supplying heroin in a deal with Allen,...**</information>**
<think> Therefore, the man who shot Chris Stockley died in 1987. **</think>**
<answer> 1987 **</answer>**

Table 6: SubSearch case study 3 (successful): SubSearch can decompose the initial information need into sequential subqueries (we need the answer of one to formulate the other) and aggregate the retrieved documents to answer correctly. However, compared to case study 2, we also see the model generated a search that is useless (see subquery 1).

Case study: failed reasoning due to missing decomposition and spurious grounding

Question: At what theater is the composer and lyricist for the musical Big Fish a residential artist?

Ground Truth: Ars Nova Theater

<think> First, I need to find the composer and lyricist of Big Fish. **</think>**
<search> what is the theater of Big Fish musical composer lyricist residential artist **</search>**
<information> Doc 1(Title: "Big Fish (musical)") Big Fish (musical) Big Fish is a musical with music and lyrics by Andrew Lippa and book by John August. It is based on Daniel Wallace's 1998 novel, """, and the 2003 film ""Big Fish"" written by John August and directed by Tim Burton. ""Big Fish"" revolves around the relationship between Edward Bloom, a travelling salesman, and his adult son Will, who looks for what is behind his father's tall stories. The story shifts between two timelines. In the present-day real world, sixty-year-old Edward Bloom faces his mortality while Will prepares to become a father himself. In the storybook past.. Doc 2(Title: "Big Fish (musical)") 'Dirty Rotten Scoundrels.' His ingratiating singing and dancing bespeak the rare man comfortable in his own skin, and he has that indefinable charismatic spark that defines a star. ""Variety"" reviewer Marilyn Stasio wrote: ""Norbert Leo Butz is cutting loose in another one of his don't-dare-miss-this perfs in 'Big Fish,' a show that speaks to anyone pining for a studiously heart-warming musical about the efforts of a dying man to justify a lifetime of lousy parenting to his alienated son."" Big Fish (musical) Big Fish is a musical with music and lyrics by Andrew Lippa and book by John August. It.. Doc 3(Title: "Big Fish: A Novel of Mythic Proportions") Big Fish: A Novel of Mythic Proportions Big Fish: A Novel of Mythic Proportions is a 1998 novel by Daniel Wallace. It was adapted into a film, ""Big Fish"", in 2003 by Tim Burton. A musical adaptation starring Norbert Leo Butz premiered in Chicago in April 2013. A young man (William Bloom), at the deathbed of his father (Edward Bloom), tries to reconcile his memories of his dad with the person he really is. Whereas he always saw his father as an irresponsible liar, he comes to understand his dad's exaggerations and their roots in reality. The book is written.. **</information>**
<think> Now I need to find where Andrew Lippa is a resident artist. **</think>**
<search> where is the theater of composer lyricist Big Fish residential artist **</search>**
<information> Doc 1(Title: "Big Fish (musical)") Big Fish (musical) Big Fish is a musical with music and lyrics by Andrew Lippa and book by John August. It is based on Daniel Wallace's 1998 novel, """, and the 2003 film ""Big Fish"" written by John August and directed by Tim Burton. ""Big Fish"" revolves around the relationship between Edward Bloom, a travelling salesman, and his adult son Will, who looks for what is behind his father's tall stories. The story shifts between two timelines. In the present-day real world, sixty-year-old Edward Bloom faces his mortality while Will prepares to become a father himself. In the storybook past.. Doc 2(Title: "Big Fish (musical)") as Karl the Giant, Brenden Lovett as Amos Calloway/Doctor Bennet, Zoe Ioannou as the Girl in the Water, Zachary Webster as the Frustrated Fisherman, and Sam Wood and Brendan Godwin sharing the role of Young Will. The production was the 12 Chairs version, produced by RPG Productions and directed by Tyran Parke. The show received its London premiere in November 2017 at The Other Palace, starring Kelsey Grammer, Jamie Muscato and Matthew Seadon-Young. Sources: PlaybillVault; Internet Broadway database ""Big Fish"" was orchestrated by Larry Hochman, with additional orchestrations by Bruce Coughlin, for a fourteen-piece orchestra. The instrumentation calls for two.. Doc 3(Title: "Andrew Lippa") shortly. The recording was released on Decca Broadway and the vocal selections are available from Hal Leonard. ""Big Fish"", Lippa's newest musical, with a book by John August, based on the 2003 film and the , premiered in Chicago, Illinois at the Oriental Theatre from April 2013 through May 5, 2013. It opened on Broadway at the Neil Simon Theatre October 6, 2013, following previews that began on September 5, 2013. Susan Stroman directed and choreographed, with scenic design by Julian Crouch, costume design by William Ivey Long and lighting design by Donald Holder. Norbert Leo Butz starred as Edward **</information>**
<think> The musical is associated with the Neil Simon Theatre, so the composer is likely a resident artist there. **</think>**
<answer> Neil Simon Theatre **</answer>**

Table 7: SubSearch case study 4 (unsuccessful): SubSearch gets stuck into generating the similar queries every time, finally reaching an answer that is not correct. While the reasoning goes well, the model fails in asking the correct 2nd question.